

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 12
 - Limits of sequences practice
 - Sequences in the real line
 - Sequences in the plane
 - Sequences in metric spaces
 - Two harder limits

The Cult of Epsilon



The Cult of Epsilon



Problem 1

Prove directly from the definition that

$$\frac{1}{n} \longrightarrow 0.$$

Solution 1

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1, \quad \text{so that } N > \frac{1}{\epsilon}.$$

For every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $1/n \rightarrow 0$.



Problem 2

Prove directly from the definition that

$$\frac{3n+1}{n+2} \longrightarrow 3.$$

Solution 2

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{5}{\epsilon} \right\rfloor + 1, \quad \text{so that } N > \frac{5}{\epsilon}.$$

For every $n \geq N$,

$$\begin{aligned} \left| \frac{3n+1}{n+2} - 3 \right| &= \left| \frac{-5}{n+2} \right| \\ &= \frac{5}{n+2} \leq \frac{5}{n} \leq \frac{5}{N} < \epsilon. \end{aligned}$$

Thus $(3n+1)/(n+2) \rightarrow 3$. □

Problem 3

Prove directly from the definition that

$$\frac{2n^2 - 1}{n^2 + 3} \longrightarrow 2.$$

Solution 3

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lceil \sqrt{\frac{7}{\epsilon}} \right\rceil + 1, \quad \text{so that } N > \sqrt{\frac{7}{\epsilon}}.$$

For every $n \geq N$,

$$\begin{aligned} \left| \frac{2n^2 - 1}{n^2 + 3} - 2 \right| &= \frac{7}{n^2 + 3} \\ &\leq \frac{7}{n^2} \leq \frac{7}{N^2} < \epsilon. \end{aligned}$$

Thus $(2n^2 - 1)/(n^2 + 3) \rightarrow 2$. □

Problem 4

Prove directly from the definition that

$$\frac{(-1)^n}{n} \longrightarrow 0.$$

Solution 4

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1.$$

For every $n \geq N$,

$$\left| \frac{(-1)^n}{n} - 0 \right| = \frac{|(-1)^n|}{n} = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Thus $(-1)^n/n \rightarrow 0$. □

Problem 5

Prove directly from the definition that

$$\left(1 + \frac{1}{n}\right)^2 \longrightarrow 1.$$

Hint: Expand the square and use $1/n^2 \leq 1/n$.

Solution 5

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{3}{\epsilon} \right\rfloor + 1.$$

For every $n \geq N$,

$$\begin{aligned} \left| \left(1 + \frac{1}{n}\right)^2 - 1 \right| &= \frac{2}{n} + \frac{1}{n^2} \\ &\leq \frac{3}{n} \leq \frac{3}{N} < \epsilon. \end{aligned}$$

Thus $(1 + 1/n)^2 \rightarrow 1$. □

Problem 6

Prove directly from the definition that

$$\sqrt{n^2 + 1} - n \longrightarrow 0.$$

Hint: Rationalize the expression.

Solution 6

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{2\epsilon} \right\rfloor + 1.$$

For every $n \geq N$,

$$\begin{aligned} \left| \sqrt{n^2 + 1} - n \right| &= \frac{(n^2 + 1) - n^2}{\sqrt{n^2 + 1} + n} \\ &\leq \frac{1}{2n} \leq \frac{1}{2N} < \epsilon. \end{aligned}$$

Thus $\sqrt{n^2 + 1} - n \rightarrow 0$. □

Problem 7

Prove directly from the definition that

$$\frac{\sin n}{\sqrt{n}} \longrightarrow 0.$$

You may use $|\sin t| \leq 1$ for every real number t .

Solution 7

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon^2} \right\rfloor + 1, \quad \text{so that } \frac{1}{\sqrt{N}} < \epsilon.$$

For every $n \geq N$,

$$\left| \frac{\sin n}{\sqrt{n}} - 0 \right| = \frac{|\sin n|}{\sqrt{n}} \leq \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \epsilon.$$

Thus $\sin n / \sqrt{n} \rightarrow 0$. □

Problem 8

Prove directly from the definition that

$$\frac{1}{2^n} \longrightarrow 0.$$

You may use $2^n \geq n + 1$ for every $n \geq 1$.

Optional: Prove this inequality by induction.

Solution 8

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1.$$

Using $2^n \geq n + 1$, for every $n \geq N$ we obtain

$$\left| \frac{1}{2^n} - 0 \right| \leq \frac{1}{n+1} < \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Thus $1/2^n \rightarrow 0$. □

For the optional induction, $2^1 = 2$. If $2^k \geq k + 1$, then

$$2^{k+1} \geq 2(k+1) \geq k+2.$$

Problem 9

Suppose a real sequence satisfies

$$|a_n - 2| \leq \frac{3}{n} \quad (n \geq 1).$$

Prove directly from the definition that

$$a_n \longrightarrow 2 \quad \text{and} \quad a_n^2 \longrightarrow 4.$$

Solution 9

Proof.

Let $\epsilon > 0$. For $n \geq 1$, the triangle inequality gives

$$|a_n + 2| \leq |a_n - 2| + 4 \leq \frac{3}{n} + 4 \leq 7.$$

Choose $N = \lfloor 21/\epsilon \rfloor + 1$. For every $n \geq N$,

$$|a_n - 2| \leq \frac{3}{n} \leq \frac{3}{N} < \epsilon,$$

$$|a_n^2 - 4| = |a_n - 2||a_n + 2| \leq \frac{21}{n} \leq \frac{21}{N} < \epsilon.$$

Thus $a_n \rightarrow 2$ and $a_n^2 \rightarrow 4$. □

Problem 10

Prove directly from the definition that

$$a_n = (-1)^n$$

does not converge in $\mathbb{R}$ with the Euclidean metric.

For every proposed limit L , find $\epsilon > 0$ such that for every positive integer N , some $n \geq N$ satisfies

$$|a_n - L| \geq \epsilon.$$

Solution 10

Proof.

Let $L \in \mathbb{R}$ be any proposed limit. Take $\epsilon = 1$. Let N be any positive integer.

If $L \leq 0$, choose the even index $n = 2N$. Then $n \geq N$ and

$$|a_n - L| = |1 - L| = 1 - L \geq 1.$$

If $L > 0$, choose the odd index $n = 2N + 1$. Then $n \geq N$ and

$$|a_n - L| = |-1 - L| = 1 + L > 1.$$

Thus no N works for $\epsilon = 1$. Since L was arbitrary, the sequence does not converge. □

Problem 11

Using Euclidean distance in $\mathbb{R}^2$, prove that

$$\left(\frac{1}{n}, \frac{2}{n}\right) \longrightarrow (0, 0).$$

Recall that

$$d((x, y), (a, b)) = \sqrt{(x - a)^2 + (y - b)^2}.$$

Solution 11

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{\sqrt{5}}{\epsilon} \right\rfloor + 1.$$

For every $n \geq N$,

$$\begin{aligned} d\left(\left(\frac{1}{n}, \frac{2}{n}\right), (0, 0)\right) &= \sqrt{\frac{1}{n^2} + \frac{4}{n^2}} \\ &= \frac{\sqrt{5}}{n} \leq \frac{\sqrt{5}}{N} < \epsilon. \end{aligned}$$

Thus $(1/n, 2/n) \rightarrow (0, 0)$. □

Problem 12

Using Euclidean distance in $\mathbb{R}^2$, prove that

$$\left(\frac{n}{n+1}, \frac{(-1)^n}{\sqrt{n}} \right) \longrightarrow (1, 0).$$

Hint: Make each coordinate error less than $\epsilon/2$.

Solution 12

Proof.

Let $\epsilon > 0$. Choose $N = \lfloor \max\{2/\epsilon, 4/\epsilon^2\} \rfloor + 1$. For every $n \geq N$,

$$\left| \frac{n}{n+1} - 1 \right| = \frac{1}{n+1} \leq \frac{1}{N} < \frac{\epsilon}{2},$$
$$\left| \frac{(-1)^n}{\sqrt{n}} \right| = \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \frac{\epsilon}{2}.$$

The Euclidean distance to $(1, 0)$ therefore satisfies

$$\sqrt{\frac{1}{(n+1)^2} + \frac{1}{n}} \leq \frac{1}{n+1} + \frac{1}{\sqrt{n}} < \epsilon.$$

Thus the sequence converges to $(1, 0)$. □

Problem 13

Using Euclidean distance in $\mathbb{R}^2$, prove that

$$\left(\frac{\cos n}{n}, \frac{\sin n}{n} \right) \longrightarrow (0, 0).$$

You may use $\cos^2 t + \sin^2 t = 1$.

Solution 13

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1.$$

For every $n \geq N$,

$$\begin{aligned} d\left(\left(\frac{\cos n}{n}, \frac{\sin n}{n}\right), (0, 0)\right) &= \sqrt{\frac{\cos^2 n + \sin^2 n}{n^2}} \\ &= \frac{1}{n} \leq \frac{1}{N} < \epsilon. \end{aligned}$$

Thus $(\cos n/n, \sin n/n) \rightarrow (0, 0)$. □

Problem 14

Equip $\mathbb{R}$ with the metric

$$d(x, y) = \sqrt{|x - y|}.$$

Prove directly that $1/n \rightarrow 0$ in this metric.

Compare your choice of $N(\epsilon)$ with your choice for the Euclidean metric in Problem 1.

Solution 14

Proof.

Let $\epsilon > 0$. Choose

$$N = \left\lfloor \frac{1}{\epsilon^2} \right\rfloor + 1.$$

For every $n \geq N$,

$$d\left(\frac{1}{n}, 0\right) = \sqrt{\left|\frac{1}{n} - 0\right|} = \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \epsilon.$$

Thus $1/n \rightarrow 0$ in this metric. □

The Euclidean proof used $N > 1/\epsilon$. This proof uses $N > 1/\epsilon^2$ because $d(1/n, 0) < \epsilon$ requires $1/n < \epsilon^2$.

Problem 15

Equip a set X with the discrete metric

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Prove that a sequence (x_n) converges to $p \in X$ if and only if it is **eventually equal to p** .

That is, there exists a positive integer N_0 such that $x_n = p$ for every $n \geq N_0$.

Solution 15

Proof.

Suppose $x_n \rightarrow p$. Take $\epsilon = 1$. By the definition of convergence, there is a positive integer N_0 such that

$$n \geq N_0 \implies d(x_n, p) < 1.$$

The discrete metric takes only the values 0 and 1, so $d(x_n, p) < 1$ forces $x_n = p$. Thus the sequence is eventually equal to p .

Conversely, suppose $x_n = p$ for every $n \geq N_0$. Let $\epsilon > 0$. Choose $N = N_0$. Then for every $n \geq N$,

$$d(x_n, p) = 0 < \epsilon.$$

Thus $x_n \rightarrow p$. □

The Cult of Epsilon returns



Problem 16

Prove directly from the definition that

$$\frac{3^n}{n!} \longrightarrow 0.$$

Hint: If $a_n = 3^n/n!$, show that

$$a_{n+1} \leq \frac{a_n}{2} \quad (n \geq 5).$$

Then compare with the geometric estimate in Problem 8.

Solution 16

Proof.

Let $\epsilon > 0$. Choose $N = 6 + \lfloor 2/\epsilon \rfloor$, so that $N \geq 6$ and $N - 5 > 2/\epsilon$.

For $n \geq N$, each factor $3/k$ with $k \geq 7$ is at most $1/2$. Since $3^6/6! = 81/80 < 2$,

$$\begin{aligned} \left| \frac{3^n}{n!} - 0 \right| &= \frac{3^6}{6!} \prod_{k=7}^n \frac{3}{k} \leq \frac{2}{2^{n-6}} \\ &\leq \frac{2}{n-5} \leq \frac{2}{N-5} < \epsilon. \end{aligned}$$

Here we used $2^m \geq m + 1$ for $m \geq 0$. Thus $3^n/n! \rightarrow 0$. □

Problem 17

Prove directly from the definition that

$$\frac{n^2}{2^n} \longrightarrow 0.$$

Hint: The binomial theorem gives

$$2^n = (1 + 1)^n \geq \binom{n}{3} = \frac{n(n-1)(n-2)}{6} \quad (n \geq 3).$$

For $n \geq 4$, compare this lower bound with a constant multiple of n^3 .

Solution 17

Proof.

Let $\epsilon > 0$. Choose $N = \max\{4, \lfloor 24/\epsilon \rfloor + 1\}$.

For $n \geq N$, we have $n - 1 \geq n/2$ and $n - 2 \geq n/2$. The binomial theorem therefore gives

$$2^n \geq \binom{n}{3} = \frac{n(n-1)(n-2)}{6} \geq \frac{n^3}{24}.$$

Consequently, for every $n \geq N$,

$$\left| \frac{n^2}{2^n} - 0 \right| \leq \frac{24}{n} \leq \frac{24}{N} < \epsilon.$$

Thus $n^2/2^n \rightarrow 0$. □